

Assignment: Git Hands-on Practice

Task 1: Configure Git

1. Set your username and email in Git.
2. Verify the configuration.

Expected Output:

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/github
$ git config --global user.name "Omjadhav13"

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/github
$ git config --global user.email "omjadhav1303@gmail.com"
```

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/github
$ git config list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symbols=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=omjadhav1303@gmail.com
user.name=Omjadhav13
```

Task 2: Initialize a Local Repository

1. Create a new folder named `git-hands-on`.
2. Initialize it as a Git repository.
3. Check the repository status.

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/github
$ git init
Initialized empty Git repository in D:/github/.git/

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ |
```

Task 3: Create and Track Files

1. Create a file named `documentation.txt`.
2. Add a short project description inside it.
3. Check file status.
4. Stage the file.
5. Commit the changes with an appropriate message.

```

Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        documentation.txt

nothing added to commit but untracked files present (use "git add" to track)

Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git add documentation.txt

Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git commit -m "documentation.txt file added"
[master (root-commit) 0018019] documentation.txt file added
 1 file changed, 1 insertion(+)
 create mode 100644 documentation.txt

Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ |

```

Task 4: Modify and Commit Changes

1. Edit `documentation.txt` and add one more line.
2. View the differences between working directory and staging area.
3. Stage and commit the changes.

```

Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git diff 0018019b429d7af052a795c537f7be4fcd0b6db9
diff --git a/documentation.txt b/documentation.txt
index 16a7f02..e9484a5 100644
--- a/documentation.txt
+++ b/documentation.txt
@@ -1,2 @@
-This is a assignment for GIT and GitHub.
\ No newline at end of file
+This is a assignment for GIT and GitHub.
+This is new added in the file.
\ No newline at end of file

```

```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git add .

Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git commit -m "New line added in the file"
[master 989f6c2] New line added in the file
1 file changed, 2 insertions(+), 1 deletion(-)
```

Task 5: View Commit History

1. View the complete commit history.
2. View commit history in one-line format.

```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git log
commit 989f6c20f5cfde51d5acfe070ff65e7a792ce2c7 (HEAD -> master)
Author: Omjadhav13 <omjadhav1303@gmail.com>
Date: Sun Dec 28 16:45:40 2025 +0530

    New line added in the file

commit 0018019b429d7af052a795c537f7be4fcd0b6db9
Author: Omjadhav13 <omjadhav1303@gmail.com>
Date: Sun Dec 28 16:30:33 2025 +0530

    documentation.txt file added
```

```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/github (master)
$ git log --pretty=oneline
989f6c20f5cfde51d5acfe070ff65e7a792ce2c7 (HEAD -> master) New line added in the file
0018019b429d7af052a795c537f7be4fcd0b6db9 documentation.txt file added
```

Task 6: Create and Work with Branches

1. Create a new branch named `feature-1`.
2. Switch to the new branch.
3. Create a new file `feature.txt`.
4. Commit the changes.

5. Switch back to the main branch.

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (master)
$ git branch feature-1

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (master)
$ git branch
  feature-1
* master

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (master)
$ git checkout feature-1
Switched to branch 'feature-1'

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (feature-1)
$ git add .

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (feature-1)
$ git commit -m "new file added in feature-1 branch"
[feature-1 f611f0c] new file added in feature-1 branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature.txt
```

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (feature-1)
$ git checkout main
Switched to branch 'main'
```

Task 7: Merge Branches

1. Merge `feature-1` branch into `main`.
2. Verify the merge using commit history.

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git merge feature-1
Already up to date.
```

```

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git log feature-1
commit f611f0cd0a129537f852190f35b82c1c6292c6b9 (HEAD -> main, feature-1)
Author: Omjadhav13 <omjadhav1303@gmail.com>
Date: Sun Dec 28 17:07:05 2025 +0530

    new file added in feature-1 branch

commit 989f6c20f5cfde51d5acfe070ff65e7a792ce2c7
Author: Omjadhav13 <omjadhav1303@gmail.com>
Date: Sun Dec 28 16:45:40 2025 +0530

    New line added in the file

commit 0018019b429d7af052a795c537f7be4fcd0b6db9
Author: Omjadhav13 <omjadhav1303@gmail.com>
Date: Sun Dec 28 16:30:33 2025 +0530

    documentation.txt file added

```

Task 8: Connect to a Remote Repository

1. Create a new repository on GitHub.
2. Add the remote repository URL to your local repo.
3. Verify the remote connection.

```

Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git remote add origin https://github.com/Omjadhav13/Git-GitHub-practice.git

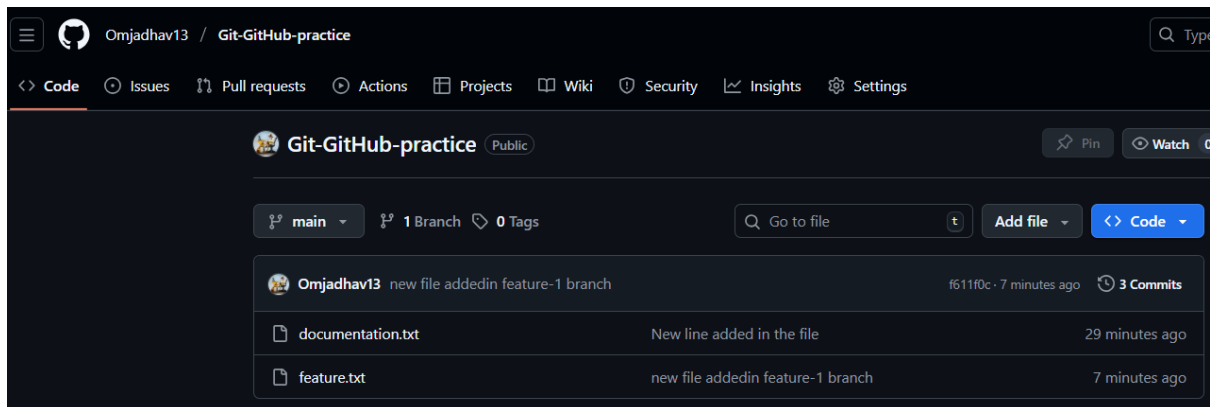
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git remote -v
origin https://github.com/Omjadhav13/Git-GitHub-practice.git (fetch)
origin https://github.com/Omjadhav13/Git-GitHub-practice.git (push)

```

Task 9: Push Code to GitHub

1. Push the `main` branch to GitHub.
2. Verify files on GitHub.

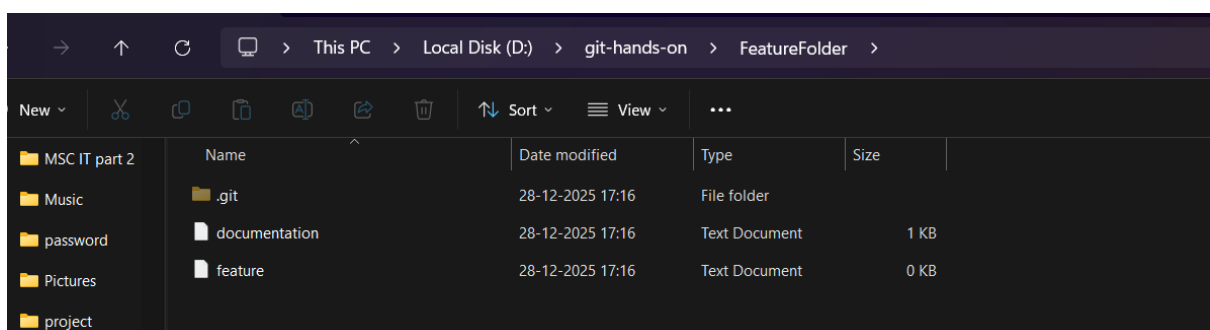
```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (9/9), 829 bytes | 829.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Omjadhav13/Git-GitHub-practice.git
* [new branch]      main -> main
```



*** **Task 10: Clone a Repository**

1. Clone the same GitHub repository into a different folder.
2. Verify the cloned content.

```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git clone https://github.com/Omjadhav13/Git-GitHub-practice.git FeatureFolder
Cloning into 'FeatureFolder'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 9 (delta 0), reused 9 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
```



Task 11: Undo Changes

1. Make a change in `documentation.txt` but do not commit.
2. Discard the changes.
3. Verify the file status.

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git diff --staged
diff --git a/documentation.txt b/documentation.txt
index e9484a5..15831f3 100644
--- a/documentation.txt
+++ b/documentation.txt
@@ -1,2 +1,3 @@
 This is a assignment for GIT and GitHub.
-This is new added in the file.
\ No newline at end of file
+This is new added in the file.
+new changes
\ No newline at end of file
```

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git restore documentation.txt
```

```
Sona1 Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   documentation.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        FeatureFolder/
```

Task 12: Remove Files9

1. Delete `feature.txt` using Git.
2. Commit the deletion.


```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git rm feature.txt
rm 'feature.txt'
```

```
Sonal Jadhav@DESKTOP-OUUESVT MINGW64 /d/git-hands-on (main)
$ git commit -m "feature.txt deleted"
[main 39ea683] feature.txt deleted
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 feature.txt
```